



Spatio-Temporal Dynamics and Factors Associated with the Coupling Coordination Between the Digital Economy and Eco-Efficiency in the Yangtze River Economic Belt

Dalai Ma¹, Kaihua Li^{2*}, Jianwei Tan¹, Helu Yang³, Jin Sun¹, Pengli Deng¹, Ruonan Chang², Yin Yan¹, Chao Hu¹

¹ School of Management, Chongqing University of Technology, 400054 Chongqing, China

² School of Economics and Finance, Chongqing University of Technology, 400054 Chongqing, China

³ School of Information and Navigation, Air Force Engineering University, 710051 Shanxi, China

* Correspondence: Kaihua Li (51240232117@stu.cqut.edu.cn)

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Abstract: Advancing the digital economy and improving eco-efficiency in a coordinated manner are important for high-quality and sustainable regional development. However, the spatio-temporal patterns of their coupling relationship and the factors associated with coordinated development remain insufficiently understood, particularly within the Yangtze River Economic Belt (YREB). To address this gap, the levels of digital economy development and eco-efficiency in 108 YREB cities were evaluated for the period 2011–2022 by integrating the entropy weight method (EWM) and the super epsilon-based measure (Super-EBM) model. The coupling coordination degree (CCD) model was subsequently employed to quantify the degree of coordinated development between the two systems, while the extreme gradient boosting (XGBoost) machine learning algorithm was applied to identify and interpret the major factors associated with the CCD. The results indicate that the digital economy development level exhibited an overall upward, albeit fluctuating, trajectory and was characterized by a distinct spatial gradient of “downstream > midstream > upstream.” In contrast, eco-efficiency remained relatively stable with moderate fluctuations and displayed the opposite spatial distribution. The CCD between the digital economy and eco-efficiency increased steadily throughout the study period. Pronounced spatial heterogeneity was observed, with consistently higher CCD values in downstream cities than in midstream and upstream areas. Improvements in the upstream and midstream regions occurred at a comparatively slower pace, which may be related to disparities in development conditions and resource endowments. Feature importance analysis based on the XGBoost model further showed that openness to the outside world, technological innovation, and urbanization generally made positive contributions to the predicted CCD, whereas government intervention and industrial structure generally made negative contributions. These findings offer new insights into the coordinated evolution of digitalization and ecological sustainability and suggest that better-targeted policy support, strengthened interregional collaboration, enhanced digital infrastructure, innovation-driven development, and deeper integration of digital technologies with green transformation should be promoted to achieve balanced and sustainable development across the YREB.

Keywords: Digital economy; Eco-efficiency; Coupling coordination degree; Machine learning; Extreme gradient boosting; Yangtze River Economic Belt

1 Introduction

With the increasingly close integration of Internet technologies, big data, and the real economy, the digital economy has moved into a new phase [1]. It now serves as an important driver of economic expansion [2]. China's 14th Five-Year Plan explicitly identifies faster digital-economy development as a national strategic priority [3]. The White Paper on the Development of China's Digital Economy [4], issued by the China Academy of Information and Communications Technology (CAICT), further indicates that digitalization has become an essential component of China's economic development. China's digital economy expanded to 50.2 trillion yuan in 2022, recording nominal

annual growth of 10.3%. Its growth rate remained above that of nominal gross domestic product (GDP) for the eleventh consecutive year. Its contribution to GDP was also close to that of the secondary sector, reaching 41.5% of the national economy. However, digital economic development remains spatially uneven across China, with pronounced polarization and substantially higher levels in the eastern region than in the western region [5]. Rapid socioeconomic expansion has intensified resource depletion and environmental pollution, increasing the conflict between economic growth and environmental protection [6], while making ecological pressures increasingly apparent. Against this background, China's transition toward high-quality growth has made green development an increasingly urgent priority [7]. Therefore, greater attention should be paid to ecological conservation and to the relationship between the digital economy and eco-efficiency, so as to promote balanced regional development.

The Yangtze River Economic Belt (YREB), one of China's major economic growth regions, faces considerable pressure to maintain economic development while strengthening ecological protection and environmental governance [8]. As a key economic corridor linking central and western China, the YREB was home to 43.1% of the country's resident population and generated 46.5% of national GDP in 2022, highlighting its substantial demographic and economic importance. However, in 2022, the region consumed 3.68 billion tons of standard coal, equivalent to 38.2% of China's total energy consumption. The region therefore faces considerable environmental pressure from high energy intensity and large carbon emissions. The digital economy can support the upgrading of manufacturing and encourage green technological innovation, thereby improving green innovation efficiency in industry [9]. However, the expansion of digital infrastructure may also create additional environmental burdens. This study uses a coupling coordination model to examine the spatial and temporal changes in the coupling coordination degree (CCD) between the digital economy and eco-efficiency and applies machine learning to identify the factors related to CCD. The findings provide empirical support for addressing the simultaneous "digital divide" and "ecological deficit" in the YREB and for advancing the strategic goal of "ecological priority and green development."

2 Literature Review

The concept of the digital economy was first proposed by Tapscott [10], who used the term "Internet economy" to describe the integration of the Internet into economic and business activities. As an emerging economic form driven by the extensive use of information technology, the digital economy has become an important force behind economic restructuring and growth. Prior research has investigated its effects in several areas, including economic growth, high-quality development [11], industrial structure [12], and employment structure [13].

Eco-efficiency refers to maximizing value while minimizing environmental pollution and resource consumption to the lowest possible levels [14]. Accurately assessing eco-efficiency is essential for research on environmental economics and sustainable development. Data envelopment analysis (DEA) and the super epsilon-based measure (Super-EBM) model are two widely used measurement tools. The DEA method has been extensively used to measure eco-efficiency [7, 15] owing to its nonparametric nature, which allows the relative efficiency of decision-making units (DMUs) with multiple inputs and outputs to be evaluated [16]. The Super-EBM model, on the other hand, is a further development of the DEA that can account for undesirable outputs [17] and thus reflect the environmental impacts of economic activities more comprehensively. Previous researchers have widely applied the Super-EBM model to efficiency evaluation [18–20]. Its application to eco-efficiency measurement reflects its importance in environmental economics research, providing a more comprehensive quantitative evaluation of eco-efficiency and creating a scientific underpinning for policy formulation and environmental management. However, much of the prior research has focused primarily on economic outputs, such as GDP or income, when measuring eco-efficiency [18–21], while giving comparatively limited attention to ecological outputs. This may hinder the identification of ecosystem weaknesses and the adoption of timely and targeted conservation measures, thereby limiting efforts to improve the overall resilience of regional ecosystems.

The relationship between the digital economy and eco-efficiency has attracted increasing attention worldwide. Researchers have applied various methods to examine this relationship and assess how digitalization is associated with eco-efficiency. Nevertheless, the pathways linking the two remain complex. Digital technologies have been reported to improve resource-use efficiency, reduce carbon emissions and environmental pollution, and support the development of eco-industries [8]. However, some studies have argued that advances in digital technology have led to the clustering of carbon-intensive industries, generating large amounts of carbon emissions [22]. Digitalization may also increase energy consumption through the expansion of the information and communication technology sector, rebound effects, and economic growth, thereby potentially exacerbating environmental pressures [23]. Attaining balanced development between economic and ecological spheres demands attention to the integration of environmental and economic benefits. Examining the coordination between the digital economy and eco-efficiency is therefore important. Most existing studies, however, investigate this relationship at relatively broad spatial scales. For example, some studies have examined China's overall eco-efficiency from a multidimensional perspective using the Super-EBM model and the spatial Durbin model [21], whereas others have investigated the CCD between China's digital economy and eco-efficiency at the provincial level [24, 25]. Therefore, this study targets the city level within the YREB region.

Moreover, existing coupling-coordination studies have mainly focused on measuring spatio-temporal patterns and regional differences [26, 27], whereas comparatively limited attention has been paid to identifying complex and nonlinear influencing mechanisms [24]. Applying machine learning methods to the analysis of these factors may help identify complex and nonlinear relationships [28]. As a branch of artificial intelligence, machine learning has the capacity to process large volumes of varied data. It can attain self-learning and self-optimization via algorithms [29] and thus more precisely capture the intricate relationships between variables as well as their dynamic evolution.

Overall, this research examines the CCD between the digital economy and eco-efficiency and its influencing factors, a topic that has attracted increasing research interest. The contributions of this study can be summarized as follows. Existing research mainly examines the CCD between the digital economy and eco-efficiency at the national or provincial level, whereas city-level studies remain limited. To fill this gap, 108 cities in the YREB are selected to assess their digital economy development and eco-efficiency. This micro-level perspective more effectively highlights the disparities in development among different regions. In addition, the Super-EBM model is applied to estimate eco-efficiency, with particular attention paid to ecological outputs by including two indicators, park green space per capita and the greening coverage rate of built-up areas. This not only reflects the concept of sustainable development but also provides another perspective for examining the coordination between the digital economy and eco-efficiency. Furthermore, the factors associated with CCD are examined using the extreme gradient boosting (XGBoost) model. The model captures nonlinear relationships and evaluates the relative importance of the selected factors, providing empirical evidence for developing region-specific optimization strategies.

3 Research Framework, Data and Methods

3.1 Research Framework

The research procedure is outlined as follows. First, data for the YREB are collected to examine the spatio-temporal changes in digital economy development and eco-efficiency. Next, the CCD model is applied to assess the coordination between the two systems. The XGBoost model is then used to identify the main factors associated with CCD. The overall analytical framework is presented in Figure 1. Finally, the findings are interpreted and discussed, and relevant recommendations are proposed to promote the coordinated development of the digital economy and eco-efficiency.

3.2 Data Sources and Indicator Construction

3.2.1 Data sources

This study covers 108 cities in the YREB over the period 2011–2022. The data were obtained from the 2012–2023 editions of the China Statistical Yearbook [30], China Energy Statistical Yearbook [31], China Statistical Yearbook on Environment [32], China City Statistical Yearbook [33]. The digital financial inclusion index was obtained from the Peking University Digital Financial Inclusion Index of China [34, 35]. Missing values were supplemented using linear interpolation.

3.2.2 Digital economy indicator system

In order to construct a comprehensive digital economy indicator system as shown in Table 1, it needs to be examined from various perspectives. Referring to prior research [7, 36], this study chooses four dimensions: digital infrastructure, scale of the digital economy, digital development potential, and level of digital finance. Digital infrastructure forms the basis for the digital economy and offers vital hardware support for its growth; the scale of the digital economy reveals its portion within the entire economy and shows how much it contributes to economic growth; the digital development potential focuses on the innovation capacity and future growth space of the digital economy; and the level of digital finance measures the extent of the application of financial technology, promoting financial innovation and efficiency improvement. These four dimensions collectively create a thorough structure for evaluating the digital economy. Seven indicators are ultimately selected to gauge the digital economy level, as presented in Table 1. These indicators include Internet penetration, mobile phone penetration, telecommunications industry development, information industry employment, innovation capacity, education investment, and digital financial inclusion. Specifically, the Peking University Digital Financial Inclusion Index serves as a proxy for digital finance development [34, 35].

3.2.3 Eco-efficiency indicator system

In order to assess eco-efficiency scientifically, this study constructs a comprehensive indicator system comprising input indicators related to capital, labor, energy, water, and land [14]; two categories of desired outputs, namely economic and ecological outputs; and undesired environmental outputs. In previous studies, economic indicators, such as GDP, have commonly been included as desired outputs in eco-efficiency measurement [21]. On this basis, this study introduces two ecological output indicators, i.e., park green space per capita and the greening coverage rate of built-up areas, to serve as desired outputs and more comprehensively reflect local ecological performance. Specific metrics are detailed in Table 2.

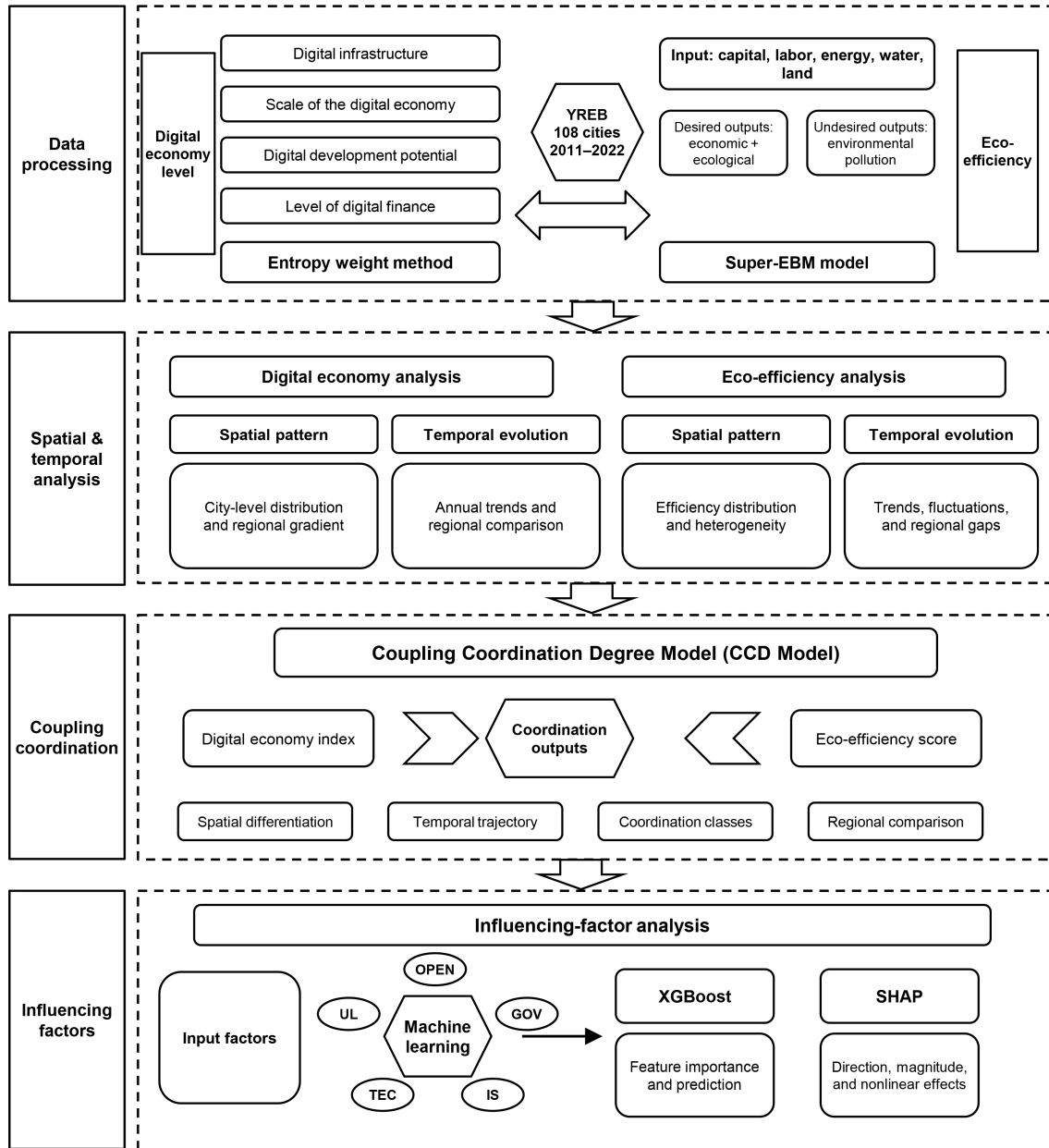


Figure 1. Research framework

Note: YREB = Yangtze River Economic Belt; Super-EBM = super epsilon-based measure; XGBoost = extreme gradient boosting; SHAP = SHapley Additive exPlanations; OPEN = openness to the outside world; GOV = government intervention; IS = industrial structure; UL = urbanization level; TEC = technological innovation.

3.2.4 Coupling influencing factors

After examining the spatial-temporal characteristics of the CCD, it remains essential to further examine the influencing factors behind it, which not only affect the synergistic development of digital economy and eco-efficiency, but can also influence the sustainability of the region. Consequently, this study selects openness to the outside world, government intervention, industrial structure, scientific and technological innovation, and urbanization level as the influencing factors. The definitions are presented in Table 3.

(i) Openness to the outside world can promote technology exchange and innovation, optimize industrial structure [37], and facilitate an improved policy environment. Meanwhile, openness to the outside world may also have complex environmental effects. For example, the introduction of pollution-intensive industries and accelerated industrialization may increase pollution, whereas export expansion and urbanization may facilitate industrial transformation and reduce environmental pressure [38, 39]. In this study, openness to the outside world is measured as the ratio of total imports and exports to regional GDP.

Table 1. Description of digital economy indicators

Dimension	Indicator	Indicator Definition	Unit	Direction
Digital infrastructure	Internet penetration	Number of Internet users per 100 households	Persons per 100 households	+
	Mobile phone penetration	Number of mobile phone subscribers at year-end per 100 persons	Subscribers per 100 persons	+
Scale of the digital economy	Telecommunications industry development	Total telecommunications services per capita	CNY per capita	+
	Information industry employment	Proportion of year-end employment in information transmission, software, and information technology services	%	+
Digital development potential	Scientific and technological support	Number of patents granted	Patents	+
	Financial support	Education expenditure	10,000 CNY	+
Level of digital finance	Digital finance development	Peking University Digital Financial Inclusion Index	Index value	+

Table 2. Description of eco-efficiency indicators

Indicator Classification	Indicator Name	Indicator Definition	Unit
Input	Capital	Fixed capital stock	10,000 CNY
	Labor	Number of employed persons at year-end	10,000 persons
	Energy input	Urban electricity consumption	10,000 kWh
	Water input	Urban water consumption	10,000 m ³
	Land input	Built-up area	km ²
Desired output	Economic output	Gross domestic product (GDP)	10,000 CNY
	Ecological output	Park green space per capita	m ²
Undesired output	Environmental pollution output	Greening coverage of built-up areas	%
		Total industrial wastewater discharge	10,000 tons
		Total industrial sulfur dioxide emissions	10,000 tons
		Total industrial solid waste generated	10,000 tons

Table 3. Description of coupling influencing factors

Indicator Name	Indicator Definition	Unit
Openness to the outside world	Total exports and imports/gross domestic product (GDP)	%
Government intervention	General government expenditure/GDP	%
Industrial structure	Value added of the secondary industry/GDP	%
Scientific and technological innovation	Expenditures on science and technology/general government expenditures	%
Urbanization level	Urban resident population/total resident population	%

(ii) The effect of government intervention may vary across cities and stages of development. For cities with relatively low levels of digital development and environmental performance, government intervention may facilitate coordinated development. However, once regional development exceeds a certain threshold, government intervention may reduce regional innovation efficiency [40] and thereby exerting a detrimental effect on coordinated development. Therefore, the effect of government intervention as a key factor needs to be considered in an integrated manner with a variety of factors and scenarios.

(iii) The secondary industry, which is generally dominated by manufacturing, is associated with relatively high energy consumption and pollution emissions [41], thereby placing considerable pressure on eco-efficiency. Although digital economy development may alleviate some of these pressures, its effects may be limited in the short term. Meanwhile, the secondary industry remains important to regional economic growth. Industrial structure is therefore measured as the ratio of the value added of the secondary industry to regional GDP [27].

(iv) Scientific and technological innovation can decrease energy costs and promote economic growth [42]. However, innovation activities may prioritize economic returns while paying insufficient attention to coordinated

economic and ecological development. In some cases, their economic benefits may not outweigh their adverse environmental effects. In this study, scientific and technological innovation is proxied by the ratio of scientific and technological innovation expenditures to general government expenditures.

(v) Urbanization provides space, infrastructure, and application scenarios for the development of the digital economy. However, previous researches have shown that urbanization may also increase carbon dioxide emissions and place pressure on the ecological environment [43–45]. In present study, urbanization is calculated as the ratio of the urban resident population to the total resident population.

3.3 Research Methods

3.3.1 Entropy weight method

The entropy weight method (EWM) assigns weights to indicators according to the variability of the observed data, thereby limiting the influence of subjective judgment. Owing to its broad applicability, it has been employed in studies of urbanization systems [46], water-quality assessment [47], and regional economic development [48]. In this study, the EWM is used to estimate the weights of the indicators included in the digital economy system. These weights are then combined through a linear aggregation procedure to calculate the composite digital economy index [24]. The detailed procedural steps are described below.

Step 1: Indicator normalization

The positive and negative indicators are as follows:

$$z_{ij,t} = \frac{x_{ij,t} - \min\{x_{ij,t}\}}{\max\{x_{ij,t}\} - \min\{x_{ij,t}\}} \quad (1)$$

$$z_{ij,t} = \frac{\max\{x_{ij,t}\} - x_{ij,t}}{\max\{x_{ij,t}\} - \min\{x_{ij,t}\}} \quad (2)$$

where, $x_{ij,t}$ is the observed value of indicator j for city i in year t , and $z_{ij,t}$ is the corresponding normalized value. The terms $\max\{x_{ij,t}\}$ and $\min\{x_{ij,t}\}$ represent the maximum and minimum of indicator j across the 108 cities in year t .

Step 2: Determining the proportion of each indicator

$$p_{ij,t} = \frac{z_{ij,t}}{\sum_{i=1}^N z_{ij,t}}, \quad 0 \leq p_{ij,t} \leq 1 \quad (3)$$

Step 3: Computing the entropy value for each indicator

$$E_{j,t} = -\frac{1}{\ln N} \sum_{i=1}^N (p_{ij,t} \ln p_{ij,t}), \quad 0 \leq E_{j,t} \leq 1 \quad (4)$$

By convention, $p_{ij,t} \ln p_{ij,t} = 0$ when $p_{ij,t} = 0$.

Step 4: Deriving the divergence coefficient for indicator j

$$G_{j,t} = 1 - E_{j,t} \quad (5)$$

Step 5: Determining the weight of indicator j

$$w_{j,t} = \frac{G_{j,t}}{\sum_{h=1}^M G_{h,t}} \quad (6)$$

Step 6: Computing the composite evaluation score

$$F_{i,t} = \sum_{j=1}^M w_{j,t} z_{ij,t} \quad (7)$$

where, $p_{ij,t}$ represents the normalized share of indicator j for city i in year t ; $E_{j,t}$ is the entropy value of indicator j ; $G_{j,t}$ is the corresponding divergence coefficient; $w_{j,t}$ is the weight assigned to indicator j ; and $F_{i,t}$ is the composite digital economy score of city i in year t . Moreover, N is the number of cities ($N = 108$), M is the number of indicators included in the digital economy system ($M = 7$), and h is the summation index over the M indicators.

3.3.2 Super epsilon-based measure model

As an enhanced non-radial efficiency assessment model, the Super-EBM model incorporates the strengths of various DEA models, including the Charnes-Cooper-Rhodes model, the Banker-Charnes-Cooper model, and the slack-based measure model. It incorporates both radial and non-radial information [17]. The model not only considers undesired outputs, but also introduces an ultra-efficiency measure [49], which enables more effective discrimination among efficient DMUs. Unlike the previous models, a parameter is introduced to indicate the share of the non-radial component in the efficiency calculation, which needs to be set before modeling [19], and takes the value in the range of [0,1]. The model is formulated as follows [20]:

$$\gamma_o^* = \min \frac{\theta - \varepsilon_x \sum_{i=1}^m \frac{w_i^- s_i^-}{x_{io}}}{\varphi + \varepsilon_y \sum_{r=1}^s \frac{w_r^+ s_r^+}{y_{ro}} + \varepsilon_b \sum_{p=1}^q \frac{w_p^{b-} s_p^{b-}}{b_{po}}} \quad (8)$$

subject to

$$\begin{cases} \sum_{\substack{j=1 \\ j \neq o}}^n x_{ij} \lambda_j + s_i^- = \theta x_{io}, & i = 1, 2, \dots, m, \\ \sum_{\substack{j=1 \\ j \neq o}}^n y_{rj} \lambda_j - s_r^+ = \varphi y_{ro}, & r = 1, 2, \dots, s, \\ \sum_{\substack{j=1 \\ j \neq o}}^n b_{pj} \lambda_j + s_p^{b-} = \varphi b_{po}, & p = 1, 2, \dots, q, \\ \lambda_j \geq 0, \quad s_i^-, s_r^+, s_p^{b-} \geq 0 \end{cases} \quad (9)$$

where, γ_o^* denotes the optimal eco-efficiency score of the evaluated DMU o ; x_{ij} , y_{rj} , and b_{pj} denote input i , desirable output r , and undesirable output p of DMU j , respectively; x_{io} , y_{ro} , and b_{po} denote the corresponding values of the evaluated DMU; s_i^- denotes the input excess; s_r^+ denotes the desirable-output shortfall; and s_p^{b-} denotes the undesirable-output excess. Furthermore, w_i^- , w_r^+ , and w_p^{b-} denote the relative weights assigned to the inputs, desirable outputs, and undesirable outputs, respectively; λ_j denotes the intensity variable; θ and φ denote the radial input and output adjustment factors, respectively; and ε_x , ε_y , and ε_b denote the parameters controlling the contributions of the non-radial components, with values ranging from 0 to 1. Finally, n , m , s , and q denote the numbers of DMUs, inputs, desirable outputs, and undesirable outputs, respectively.

3.3.3 Coupling coordination degree model

This study investigates the degree of coordination between the digital economy and eco-efficiency systems. Their coupling level is measured using Eq. (10).

$$C = 2 \left[\frac{f(DEL) g(EE)}{(f(DEL) + g(EE))^2} \right]^{\frac{1}{2}} \quad (10)$$

The coupling degree captures the intensity of interaction between the two systems. However, a high coupling value may also arise when both systems remain at similarly low development levels, thereby creating pseudo-coordination. To address this limitation, the CCD model is employed to consider both interaction intensity and overall development. The CCD and the composite development index are determined as follows:

$$T = \alpha f(DEL) + \beta g(EE) \quad (11)$$

$$D = \sqrt{C \times T} \quad (12)$$

where, $f(DEL)$ and $g(EE)$ denote the comprehensive evaluation scores of the digital economy and eco-efficiency, respectively; C stands for the coupling degree; T signifies the comprehensive development index; and α and β denote the contribution coefficients of the two systems, satisfying $\alpha + \beta = 1$. In high-quality economic development, digital economy is as important as eco-efficiency development, so α and β are taken as 0.5 [7, 24]; D denotes the CCD and ranges from 0 to 1, with a larger value indicating a higher level of coordinated development between the two systems. The CCD, based on related studies [46], is categorized into four intervals, which are shown in Table 4.

Table 4. Classification criteria for the coupling coordination degree (CCD)

CCD Value	Classification
$0 \leq D \leq 0.4$	No coordination
$0.4 < D \leq 0.5$	Low coordination
$0.5 < D \leq 0.7$	Medium coordination
$0.7 < D \leq 1.0$	High coordination

Note: D denotes the CCD and ranges from 0 to 1.

3.3.4 Extreme gradient boosting model

Gradient boosting decision trees (GBDT) constitute an ensemble learning method in which multiple decision trees are constructed sequentially to improve predictive performance. XGBoost extends the conventional GBDT framework by introducing second-order Taylor expansion of the loss function and regularization terms that help control model complexity and reduce overfitting [50]. Owing to its predictive accuracy, computational efficiency, and flexible parameter configuration, XGBoost has been applied extensively across various research fields [28, 51]. The objective function, $\text{Obj}(\Theta)$, combines the training loss with a penalty for model complexity:

$$\text{Obj}(\Theta) = \sum_{i=1}^{N_{\text{obs}}} l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (13)$$

where, N_{obs} denotes the total number of observations; y_i and $\hat{y}_i$ are the observed and predicted values for observation i , respectively; $l(\cdot)$ denotes the loss function; K is the total number of decision trees; f_k denotes the k -th decision tree; $\Omega(f_k)$ is the regularization term used to constrain model complexity; and Θ denotes the set of model parameters.

Model performance was assessed using three metrics: coefficient of determination (R^2), root mean square error (RMSE) and mean absolute error (MAE), which are expressed as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{N_{\text{obs}}} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N_{\text{obs}}} (y_i - \bar{y})^2} \quad (14)$$

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{obs}}} \sum_{i=1}^{N_{\text{obs}}} (y_i - \hat{y}_i)^2} \quad (15)$$

$$\text{MAE} = \frac{1}{N_{\text{obs}}} \sum_{i=1}^{N_{\text{obs}}} |y_i - \hat{y}_i| \quad (16)$$

where, $\bar{y}$ denotes the mean of the observed values. Better predictive performance is indicated by a higher R^2 and smaller RMSE and MAE values.

3.3.5 Shapley Additive exPlanations method

The Shapley Additive exPlanations (SHAP) method, developed on the basis of Shapley values, assigns a contribution value to each feature for an individual model prediction [52, 53]. For a given observation, a positive SHAP value indicates that the corresponding feature shifts the model output above the baseline, whereas a negative SHAP value shifts it below the baseline. Accordingly, the model prediction for the input feature vector $\mathbf{x}$ can be decomposed as follows:

$$f(\mathbf{x}) = \phi_0 + \sum_{\ell=1}^d \phi_{\ell} \quad (17)$$

where, $f(\mathbf{x})$ denotes the model prediction for the input feature vector $\mathbf{x}$; ϕ_0 denotes the expected model output over the background dataset; ϕ_{ℓ} denotes the SHAP value of feature ℓ ; and d denotes the total number of input features.

The SHAP value of feature ℓ is calculated as follows:

$$\phi_{\ell} = \sum_{S \subseteq \mathcal{F} \setminus \{\ell\}} \frac{|S|! (d - |S| - 1)!}{d!} [f(\mathbf{x}_{S \cup \{\ell\}}) - f(\mathbf{x}_S)] \quad (18)$$

where, $\mathcal{F}$ denotes the full set of input features; $S \subseteq \mathcal{F} \setminus \{\ell\}$ is a subset of features excluding feature ℓ ; and $|S|$ denotes the number of features contained in S . The terms $\mathbf{x}_S$ and $\mathbf{x}_{S \cup \{\ell\}}$ denote the model inputs formed without and with feature ℓ , respectively. Moreover, $\frac{|S|!(d-|S|-1)!}{d!}$ is the weighting factor associated with each possible feature ordering, while $f(\mathbf{x}_{S \cup \{\ell\}}) - f(\mathbf{x}_S)$ measures the change in the model prediction resulting from the inclusion of feature ℓ in subset S .

4 Results

4.1 Digital Economy Analysis of the Yangtze River Economic Belt

4.1.1 Temporal evolution of the digital economy in the Yangtze River Economic Belt

Figure 2 shows the mean level of digital economy development for 108 YREB cities, together with the corresponding values for the upstream, midstream, and downstream regions, over the period 2011–2022. The level of digital economy development increased across all three regions, with a persistent spatial pattern of “downstream > midstream > upstream.” For the YREB as a whole, the average digital economy development level increased from 0.0583 in 2011 to 0.1471 in 2022, showing an overall upward trend with moderate fluctuations.

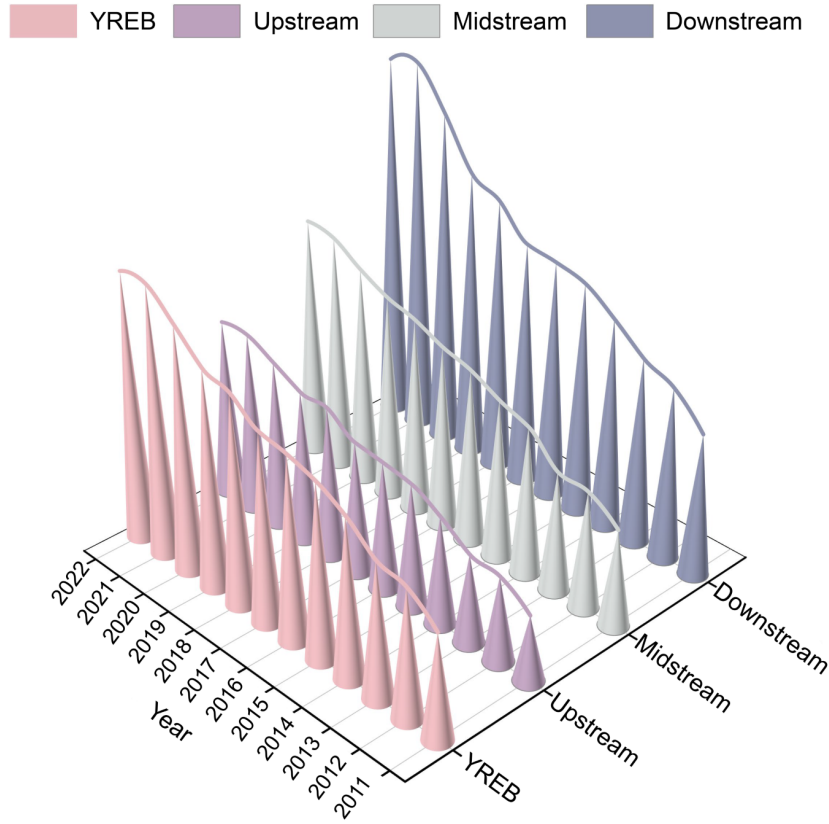


Figure 2. Temporal evolution of the average digital economy development level in the Yangtze River Economic Belt (YREB) and its upstream, midstream, downstream regions (2011–2022)

At the regional level, the mean values for 2011–2022 were 0.1398 in the downstream region, 0.0958 in the midstream region, and 0.0716 in the upstream region. The differences among the three regions are relatively pronounced. The digital economy development level in the upstream region is much lower than the overall average, while that in the midstream region is slightly lower. By comparison, the downstream region has a considerably higher digital economy development level than the overall average. Nevertheless, all three regions show an increasing trend. This suggests strong growth momentum in the YREB amid continued policy support and infrastructure development. Although regional differences remain, the overall development trend is positive.

4.1.2 Spatial characterization of digital economy in the Yangtze River Economic Belt

In this study, the results for 2011, 2015, 2019, and 2022 were selected for visualization in Figure 3 to provide a clearer understanding of the spatial evolution of the digital economy in the YREB. As shown in Figure 3, the digital economy exhibits pronounced spatial heterogeneity. The number of cities with very high digital economy development levels is relatively small. These cities include Shanghai, Suzhou, Hangzhou, and Chongqing and are

mainly located in the downstream region or are municipalities directly under the central government. Better-developed cities are mainly concentrated in the downstream region, as well as in provincial capitals such as Chengdu, Wuhan, and Changsha in the upstream and midstream regions. Cities with relatively low digital economy development levels, including Pu'er, Yuxi, Baoshan, Lijiang, and Zhaotong, are mainly concentrated in the upstream region. Most cities exhibit moderate digital economy development levels and are distributed throughout the upstream and midstream regions and the northern part of the downstream region. Cities in the downstream Yangtze River Delta generally have stronger digital infrastructure and a more developed digital-industry base, whereas digital economy development in the midstream and upstream regions remains comparatively [8, 54].

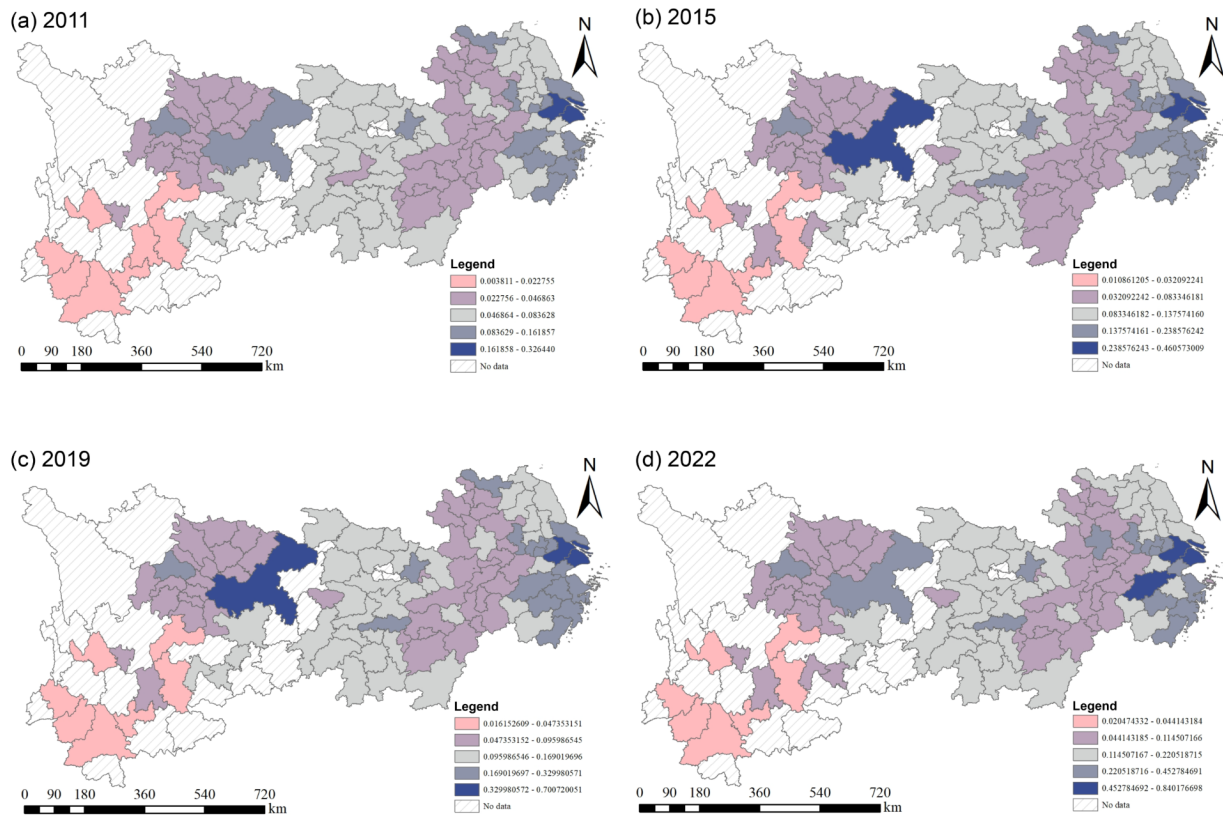


Figure 3. Spatial evolution of digital economy development level in (a) 2011, (b) 2015, (c) 2019, and (d) 2022

Overall, the digital economy exhibits a spatial pattern of “downstream > midstream > upstream.” This pattern may be associated with the downstream region’s stronger economic foundation, more complete infrastructure, more advanced industrial structure, and larger pool of skilled professionals. Cities such as Shanghai, Hangzhou, Nanjing, and Suzhou are among the most economically active cities in China, and industries related to the digital economy, including the internet and financial technology, developed relatively early in these cities. The midstream region, including Wuhan, Changsha, and Nanchang, has a solid economic foundation, but its industrial structure remains largely dominated by traditional manufacturing. Consequently, the application of digital technologies remains comparatively limited, and the overall pace of development is relatively slow. The upstream region includes Chongqing and cities in Yunnan, Guizhou, and Sichuan. Although Chongqing and the provincial capitals of Chengdu, Kunming, and Guiyang have relatively strong economic foundations, weaker economic development in many other upstream cities constrains the overall development of the digital economy.

4.2 Eco-Efficiency Analysis of the Yangtze River Economic Belt

4.2.1 Temporal characterization of eco-efficiency in Yangtze River Economic Belt

Figure 4 shows the average eco-efficiency of the 108 cities in the YREB from 2011 to 2022. Overall, eco-efficiency exhibits a fluctuating pattern of “rise–fall–rise–fall–rise,” forming a slight W-shaped trajectory from 2016 to 2022. As shown in Figure 5, the eco-efficiency levels and trends across the three regions exhibit certain regional disparities. The upstream region follows a trend most similar to the overall YREB average during 2011–2022. From 2011 to 2012, the differences between the overall YREB average and the averages of the upstream, midstream, and downstream regions were relatively small. From 2012 to 2016, the interregional differences gradually widened, with the eco-efficiency of the downstream region remaining below the overall YREB average. The downstream region contains many highly

industrialized and urbanized cities, where resource consumption and environmental pollution pressures are relatively prominent. From 2016 to 2018, the differences among the regional eco-efficiency averages narrowed. This period followed a major policy shift in the YREB, marked by the call at a national symposium in early 2016 to promote well-coordinated environmental conservation and avoid excessive development [55]. From 2018 to 2022, regional disparities widened again. On this period, the upstream region generally improved, whereas the midstream region declined after 2019 and the downstream region fluctuated considerably, reaching a low point in 2021 before recovering in 2022.

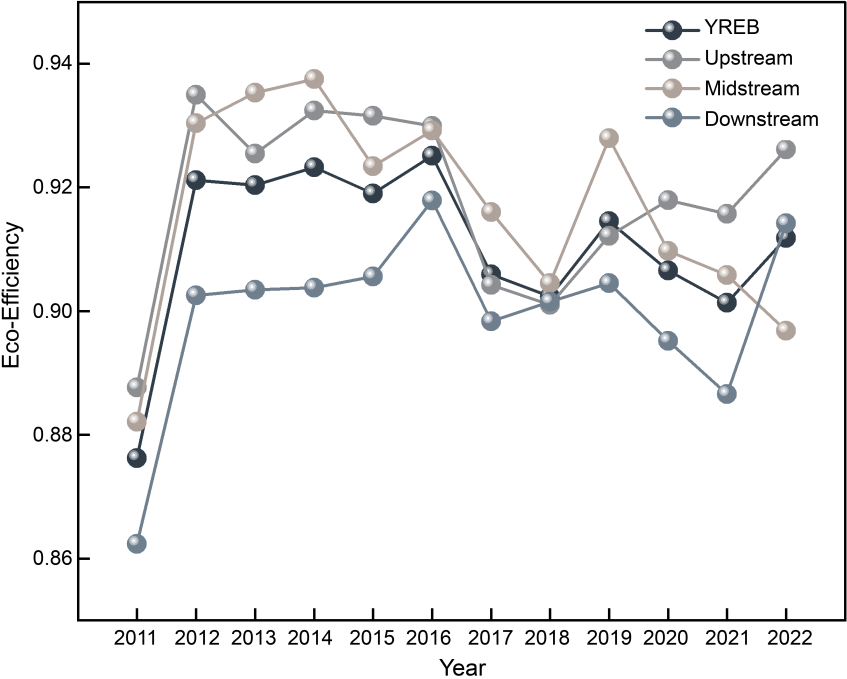


Figure 4. Comparative temporal evolution of eco-efficiency in the Yangtze River Economic Belt (YREB) and its upstream, midstream, and downstream regions from 2011 to 2022

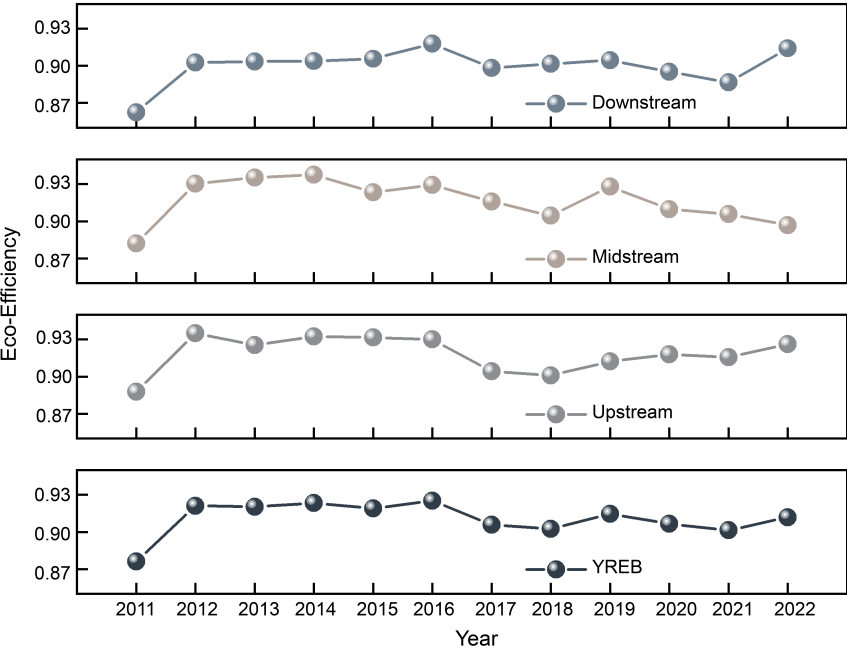


Figure 5. Separate temporal trends in eco-efficiency for the Yangtze River Economic Belt (YREB) and its downstream, midstream, and upstream regions from 2011 to 2022

4.2.2 Spatial characterization of eco-efficiency in the Yangtze River Economic Belt

The eco-efficiency results for 2011, 2015, 2019, and 2022 were visualized to facilitate the observation of the spatial evolution of eco-efficiency in the YREB, as shown in Figure 6. It is evident that eco-efficiency exhibits spatial heterogeneity across the YREB. In 2011, cities with relatively low eco-efficiency were mainly concentrated in the midstream and downstream regions, including Xuzhou, Suqian, and Zhoushan, as well as Chongqing in the upstream region. This may be related to the relatively high proportion of secondary and heavy chemical industries, which placed considerable pressure on the ecological environment. In 2015, eco-efficiency improved in some upstream and midstream cities, including Qujing, Ya'an, Mianyang, and Changde. In the downstream region, although eco-efficiency also improved in cities such as Taizhou, Suzhou, Huzhou, and Xuancheng, some cities continued to exhibit relatively low levels. In 2019, the overall eco-efficiency of the YREB was in a stage of gradual recovery. Although the overall level remained lower than that in 2015, cities in the midstream and downstream regions showed relatively rapid improvements in eco-efficiency. In 2022, eco-efficiency improved more rapidly in the upstream and midstream regions, whereas improvement in the downstream region was relatively slow, and nearly half of the cities still exhibited relatively low eco-efficiency levels. Upstream regions implemented relatively strict ecological red-line protection policies, restricted the entry of highly polluting industries, and invested substantially in ecological restoration projects. By contrast, the long-term policy orientation of downstream regions placed greater emphasis on industrial agglomeration and economic output, while the intensity of ecological governance was relatively limited before 2017. Midstream cities faced trade-offs between industrial growth and ecological protection, resulting in moderate levels of eco-efficiency.

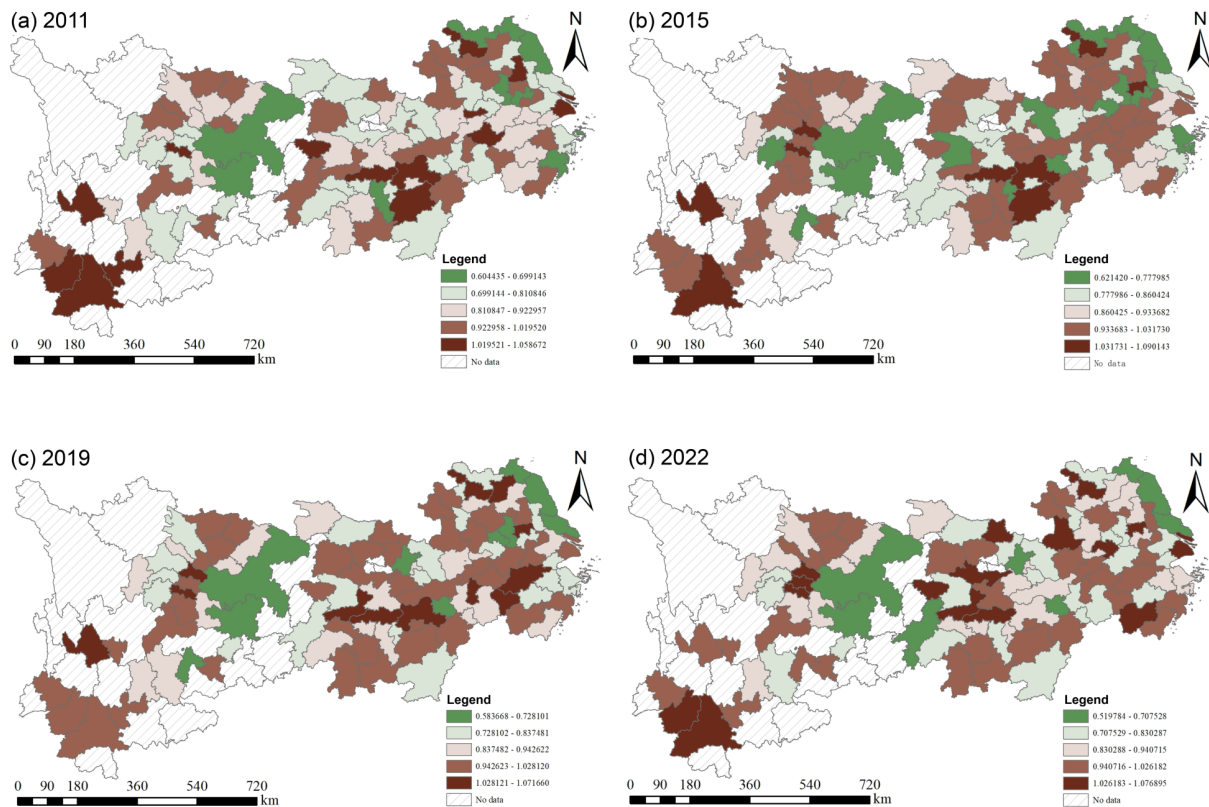


Figure 6. Spatial evolution of eco-efficiency in the Yangtze River Economic Belt (YREB) in (a) 2011, (b) 2015, (c) 2019, and (d) 2022

Note: Areas shown in white represent cities outside the study sample or areas for which data were unavailable.

In general, regional disparities in eco-efficiency exist throughout the YREB, showing an overall pattern of “upstream > midstream > downstream.” This pattern may partly be related to the inclusion of two ecological indicators—park green space per capita and the greening coverage of built-up areas—as desirable output indicators in the eco-efficiency assessment, which places greater emphasis on ecological performance. The relatively high eco-efficiency of the upstream region may result from its favorable ecological and geographical conditions, together with the emphasis placed by local governments on ecological protection through large-scale ecological restoration and greening projects. The midstream region has the second-highest eco-efficiency level. Although cities such as Wuhan and Changsha have made progress in ecological development while experiencing rapid economic growth, the region

upstream region is lower. The CCD in the midstream region shows the smallest difference from the overall average and follows the most similar temporal trend. These differences demonstrate the uneven development of coupling coordination across the YREB. In the temporal dimension, the downstream region, with its strong economic foundation and preferential policies, started to emphasize the synergistic enhancement of digital economy and eco-efficiency earlier, and achieved considerable progress during the initial stage. The midstream region also made steady progress but developed more slowly than the downstream region. Due to the weak economic foundation of the upstream region and the greater challenges faced in developing the digital economy, the coordinated development of the two systems was more difficult to achieve than in the other regions. Consequently, the upstream region exhibited a relatively slow rate of improvement throughout the study period, reflecting clear temporal disparities compared with the midstream and downstream regions.

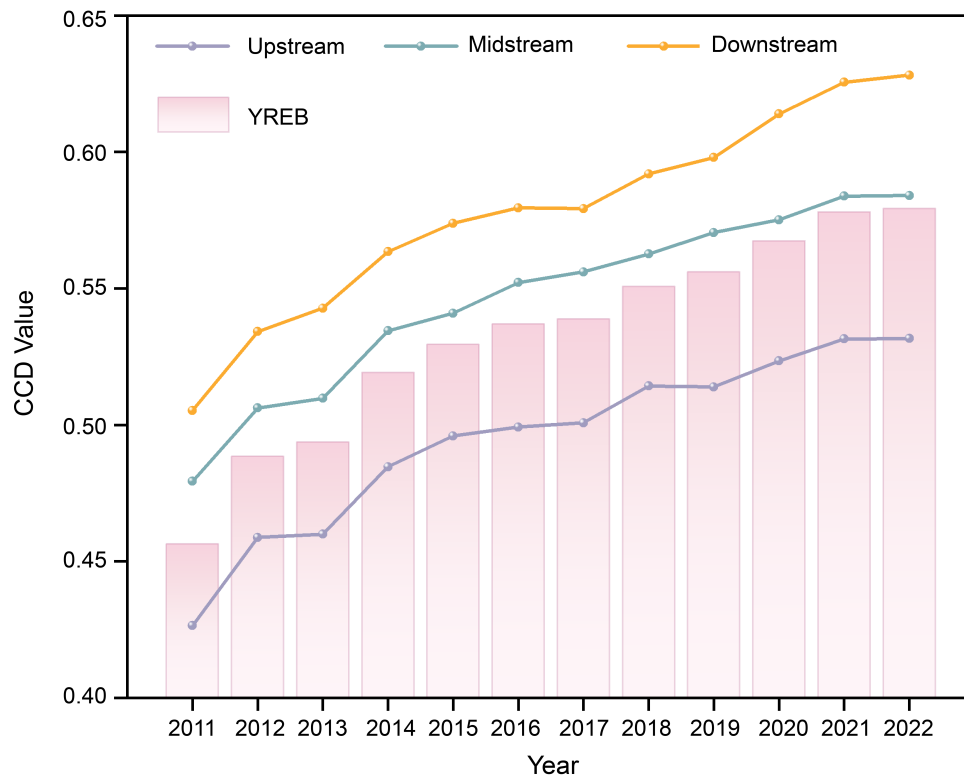


Figure 8. Temporal evolution of the average coupling coordination degree (CCD) in the Yangtze River Economic Belt (YREB) and its upstream, midstream, and downstream regions (2011–2022)

4.3.2 Spatial characterization of the coupling coordination degree in the Yangtze River Economic Belt

The CCD between the digital economy and eco-efficiency exhibits distinct geographical patterns across the YREB. To further examine its spatial variation, four representative years—2011, 2015, 2019, and 2022—were selected, as shown in Figure 9 and Figure 10. Overall, the CCD levels of the 108 cities increased throughout the study period. In 2011, cities with relatively high CCD values, including Shanghai, Ningbo, Hangzhou, and Suzhou, were mainly concentrated in the downstream region. As also shown in Figure 7, these cities maintained comparatively high CCD values throughout the study period and were concentrated in the economically developed areas of Jiangsu, Zhejiang, and Shanghai. By contrast, cities with relatively low CCD values, including Pu'er, Qujing, Baoshan, and Lijiang, were mainly concentrated in the upstream region. Other upstream cities, such as Mianyang, Guangyuan, Dazhou, and Guang'an, also exhibited relatively low CCD values. In 2015, the proportion of cities in the low-coordination category decreased from 68% to 31%, while the proportion in the medium-coordination category increased from 19% to 62%, indicating a substantial overall improvement. Spatially, several downstream cities, including Wenzhou, Hefei, and Huai'an, exhibited increased CCD values. Chongqing and Chengdu in the upstream region also showed relatively rapid improvements, which may be associated with regional development policies, e.g. "Chengdu-Chongqing Economic Zone," and increased emphasis on technological innovation and sustainable development. In 2019, the CCD continued to increase, and the proportion of cities in the medium-coordination category rose from 62% to 77%. Midstream cities such as Jingmen, Yueyang, Changsha, and Ganzhou also showed marked improvements. These changes may be associated with industrial restructuring, the transition toward technology-intensive and low-carbon industries, and enhanced regional coordination among urban agglomerations. By 2022, the proportion of cities in the

medium-coordination category had increased to 80%, while downstream cities generally maintained higher CCD levels.



Figure 9. Distribution of cities across coupling coordination degree (CCD) categories in 2011, 2015, 2019, and 2022

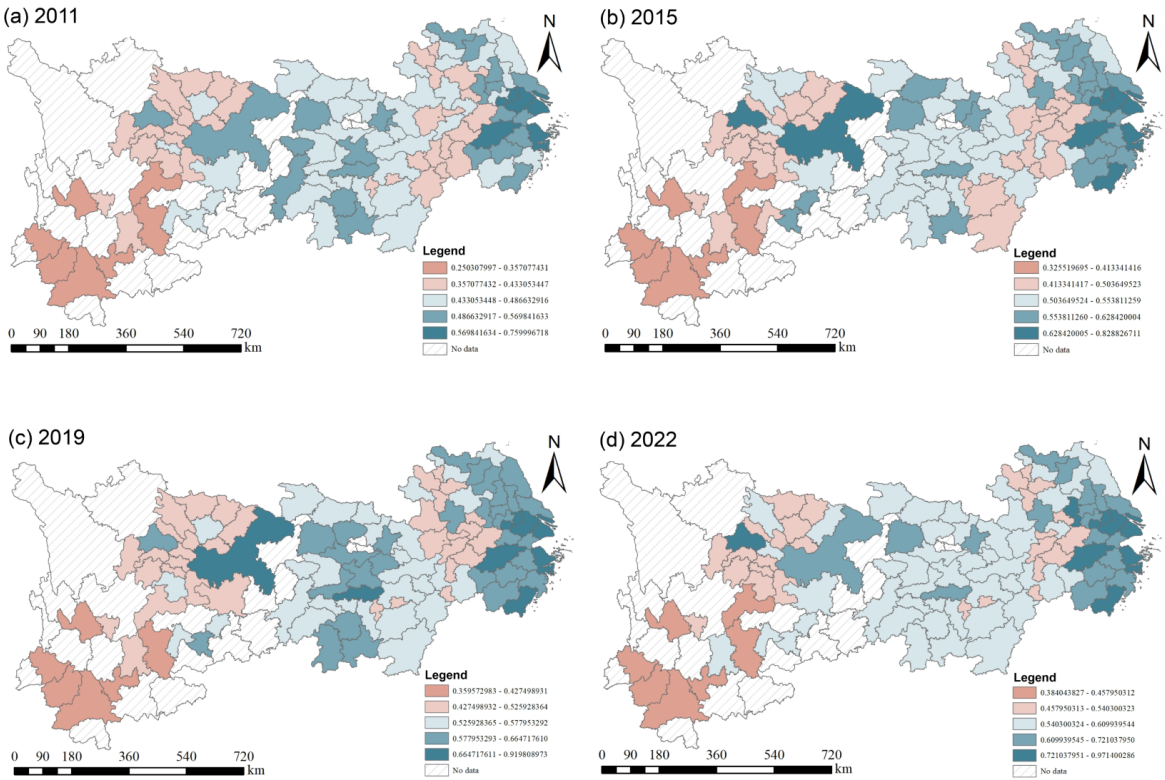


Figure 10. Spatial distribution of the coupling coordination degree (CCD) in the Yangtze River Economic Belt (YREB) in (a) 2011, (b) 2015, (c) 2019, and (d) 2022

The CCD between digital economy and eco-efficiency in YREB shows an overall spatial pattern of “downstream > midstream > upstream.” This disparity may stem from a mix of factors like regional economic conditions, industrial composition, support, technological innovation, and resource allocation. The downstream region, particularly the Yangtze River Delta, has a strong economic foundation and relatively well-developed digital infrastructure, industrial digitalization, and environmental governance. These conditions support the coordinated development of the digital economy and eco-efficiency. Although the midstream region has a certain foundation for economic development, there are gaps in technological innovation and eco-governance compared to the downstream region. The midstream region’s industrial structure shows relatively high dependence on traditional industries, and may also constrain improvements in CCD. The upstream region faces greater challenges because of its comparatively weaker economic foundation, less-developed digital infrastructure, limited technological innovation capacity, and shortages or outflows of skilled professionals. Downstream cities enjoyed overlapping policy dividends including digital economy pilot, ecological civilization demonstration zone and integrated regional development, with sufficient fiscal revenue to support synchronous digital construction and environmental governance; midstream cities have single fiscal sources, policy implementation focuses on either industrial growth or ecological governance with insufficient balanced investment; upstream cities face fiscal constraints, ecological protection fiscal expenditure squeezes digital infrastructure investment, and digital policy implementation lacks special financial support, forming a dual restriction on coupling coordination promotion.

To summarize, the regional heterogeneity of CCD reflects interregional imbalances in economic growth level, industrial composition, technological innovation ability and resource allocation. Narrowing the interregional digital economy development gap, optimizing industrial composition, enhancing technological innovation capacity and strengthening the governance of the ecological environment are the keys to achieving the overall coordinated development of YREB.

4.4 Coupling Coordination Degree Influencing Factor Analysis of the Yangtze River Economic Belt

4.4.1 Model testing

The XGBoost model was adopted to analyze factors influencing the CCD [50]. Through this model, the relative contributions of openness to the outside world, government intervention, industrial structure, technological innovation, and the level of urbanization factors on the coupled coordination relationship are discussed in depth. RMSE, R^2 , and MAE were used to evaluate model performance. Smaller RMSE and MAE values indicate better predictive performance [56], whereas values of R^2 approaching 1 reflect the model’s ability to account for a larger share of the variance in the dependent variable [57].

The model-fitting results for 2011, 2015, 2019, and 2022 are presented in Table 5. The training-set RMSE values for these four years were 0.0013, 0.0025, 0.0149, and 0.0010, respectively. The corresponding test-set RMSE values were 0.0116, 0.0083, 0.0140, and 0.0078, respectively. The relatively small RMSE values indicate that the model achieved high predictive accuracy. For both the training and test sets, the R^2 values remained above 0.9000 in all four years, demonstrating a strong model fit. Similarly, the MAE values remained small across both datasets. Overall, the XGBoost model demonstrated good fitting and predictive performance and was therefore considered suitable for analyzing the factors influencing the CCD.

Table 5. Extreme gradient boosting (XGBoost) model fitting results

Year	Dataset	Root Mean Square Error (RMSE)	R^2	Mean Absolute Error (MAE)
2011	Training set	0.0013	0.9996	0.0010
	Test set	0.0116	0.9602	0.0097
2015	Training set	0.0025	0.9986	0.0019
	Test set	0.0083	0.9719	0.0069
2019	Training set	0.0149	0.9621	0.0107
	Test set	0.0140	0.9311	0.0104
2022	Training set	0.0010	0.9999	0.0007
	Test set	0.0078	0.9931	0.0059

4.4.2 Analysis of influencing factors

For a deeper exploration of the factors influencing the CCD, SHAP summary plots of each influencing factor for 2011, 2015, 2019, and 2022 are presented in Figure 11. From the scattered point distribution in Figure 11, high values of urbanization and scientific and technological innovation are generally associated with positive SHAP contributions, while high values of government intervention generally show negative SHAP contributions, consistent with the model results. Each point represents a city, and its horizontal position represents the contribution of the corresponding

feature to the model prediction. The more dispersed the points are along the horizontal axis, the greater the variation in the contribution of the feature across cities. If the points with high feature values are predominantly distributed on the right side, high values of the feature tend to increase the predicted CCD, and vice versa, high feature values distributed on the left tend to decrease the predicted CCD. The zero line indicates that the feature makes little or no contribution to the prediction relative to the baseline value for a specific observation. However, points close to zero do not necessarily indicate that the feature has a minor overall influence on the model.

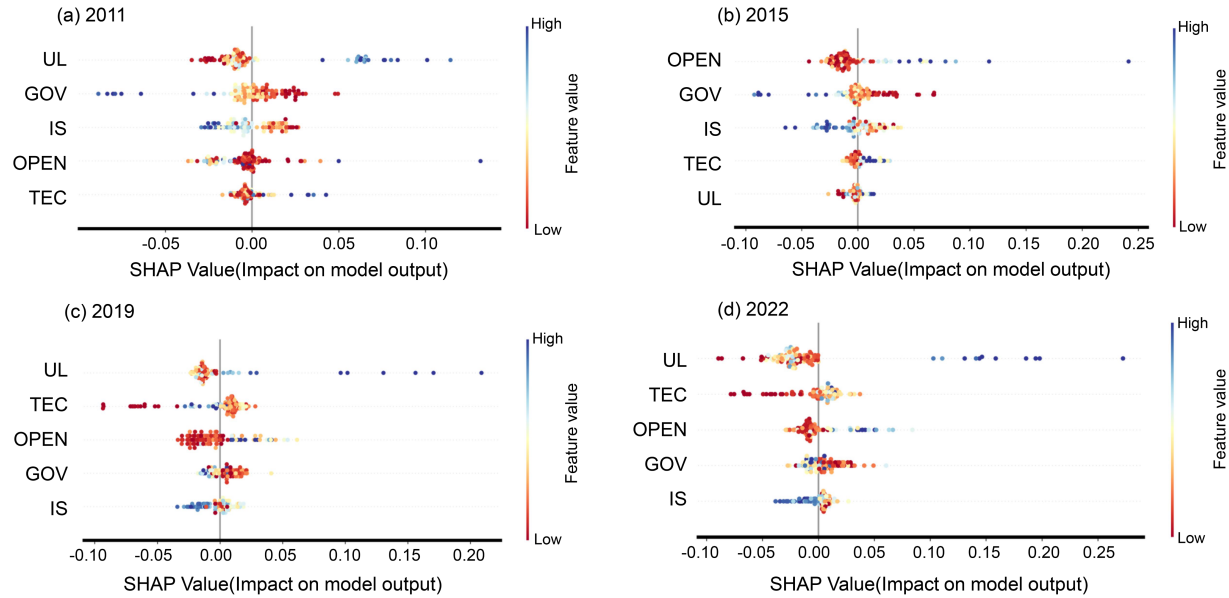


Figure 11. SHAP summary plots of five factors influencing the CCD in (a) 2011, (b) 2015, (c) 2019, and (d) 2022

Note: SHAP = SHapley Additive exPlanations; CCD = coupling coordination degree; OPEN = openness to the outside world;

GOV = government intervention; IS = industrial structure; UL = urbanization level; TEC = technological innovation.

The degree of openness to the outside world generally makes a positive contribution to the predicted CCD, although its contribution varies across years. The largest positive SHAP values for openness were observed in 2015, while its contribution was comparatively weaker in 2019 and 2022. One possible explanation is that local governments absorb foreign investment to introduce international digital enterprises and projects, which can boost the local digital economy's development. Simultaneously, a high degree of openness may facilitate the introduction of advanced technologies and high-quality foreign investment, thereby promoting green technological innovation and reducing carbon emissions [37]. After 2015, the global economic situation became increasingly complex and volatile, trade protectionism increased, and international trade frictions intensified. These changes in the external environment may have a greater influence on China's export-oriented economy, which in turn may affect the role of the degree of openness to the outside world in promoting CCD. Overall, these results highlight a time-varying association between openness and CCD at the city level.

The degree of government intervention generally makes a negative contribution to the predicted CCD, although its relative importance varies across the four years. Government intervention may inhibit the region's innovation efficiency [58] and weaken enterprises' incentives to innovate, particularly when enterprises become more dependent on policy support rather than proactively investing resources in the digital economy and environmentally friendly technologies. Moreover, the market mechanism may help identify the synergistic potential between digital economy and eco-efficiency, excessive government intervention may restrict this mechanism to some extent, which in turn may affect the CCD. This interpretation is broadly consistent with Chen and Bu [59], who found that political connections may encourage rent-seeking and divert resources away from corporate green innovation. Overall, the results indicate a negative model-based association between government intervention and CCD.

Industrial structure makes a relatively weak and generally negative contribution to the predicted CCD. This may reflect the energy-intensive nature of the secondary industry, which includes manufacturing, mining, and construction. A larger share of secondary industry has been linked to more severe urban air pollution [41]. Although digital economy development may ease some of these pressures, it may not fully compensate for the adverse effects on eco-efficiency at the current development stage. Previous research indicates that the digital economy can improve urban green economic efficiency through technological innovation and changes in industrial and energy structures [60]. However, its effect on industrial carbon emission efficiency may be nonlinear and depend on the level of green technological innovation [61]. Therefore, the negative SHAP contribution observed in this study suggests that the digital and

green transformation of the secondary industry may not yet be sufficient to fully offset the resource consumption and environmental pressures associated with industrial production at the city level.

Scientific and technological innovation is generally associated with higher predicted CCD values, although its relative importance changes over time. Technological innovation supports the development and application of digital technologies. Through artificial intelligence, blockchain, and other technologies that improve resource allocation, the digital economy may also encourage innovation in green production methods [62]. Digital economy development can further improve urban eco-efficiency by reducing resource misallocation and advancing green technologies [63], thus supporting coordination between the digital economy and eco-efficiency. Moreover, government support policies, such as research and development grants and tax concessions for scientific and technological innovation, can also stimulate enterprise innovation and create favorable conditions for the coordinated development of the two systems [64]. This result is broadly consistent with Deng et al. [62], who reported that the digital economy promotes green productivity in manufacturing. The XGBoost–SHAP analysis used in this study further evaluates the contribution of technological innovation to CCD at the municipal level.

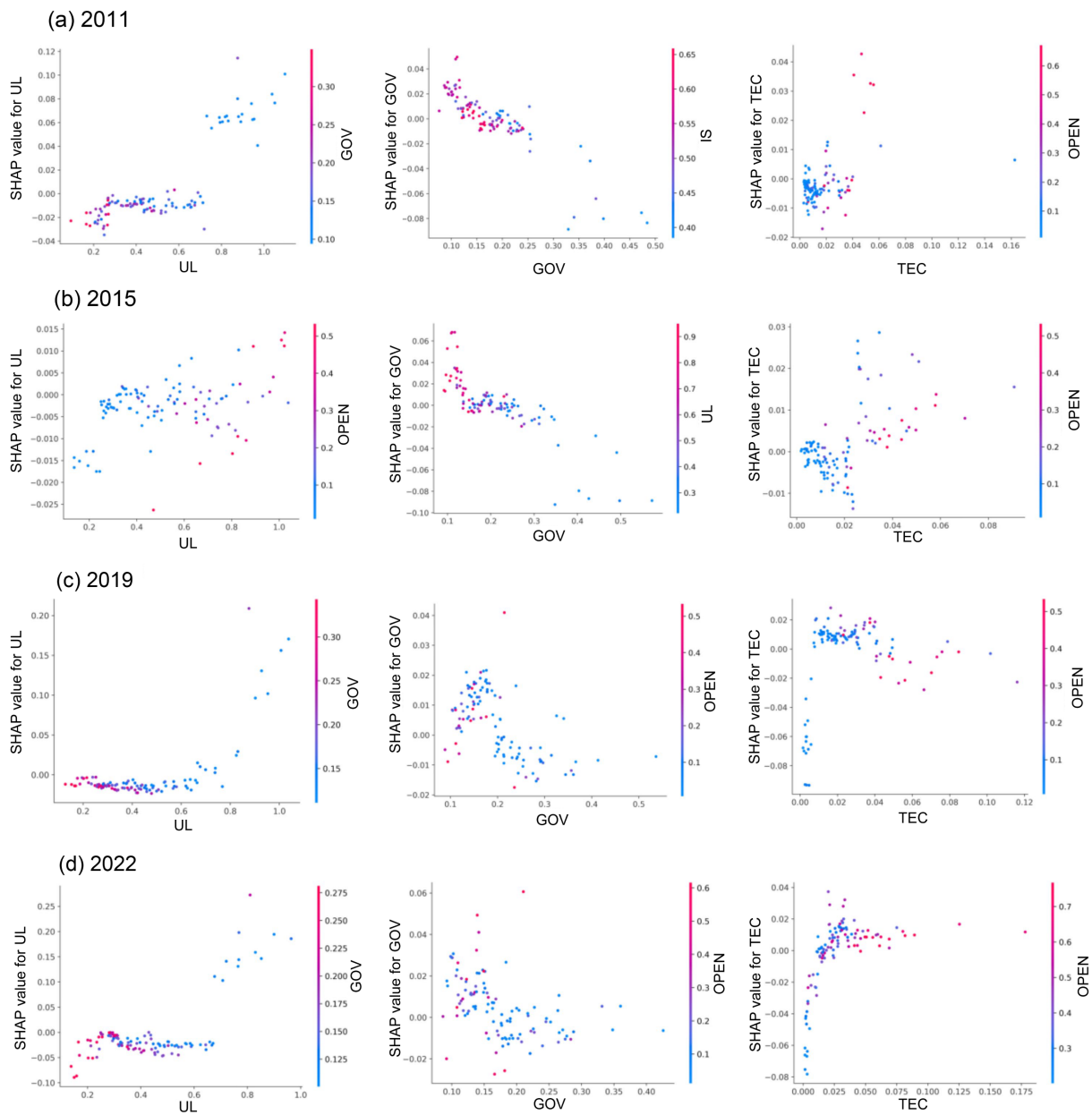


Figure 12. SHAP feature dependence plots for urbanization, government intervention, and scientific and technological innovation in (a) 2011, (b) 2015, (c) 2019, and (d) 2022

Note: SHAP = SHapley Additive exPlanations; OPEN = openness to the outside world; GOV = government intervention; IS = industrial structure; UL = urbanization level; TEC = technological innovation.

Urbanization is generally associated with higher predicted CCD values and was the most influential factor in 2011, 2019, and 2022, whereas openness to the outside world ranked first in 2015. Urbanization can facilitate the concentration and efficient allocation of resources, promote the development of the digital economy, and improve infrastructure by strengthening the interaction among knowledge, industrial structures, and resource allocation [65]. New urbanization can also help reduce pollution emissions and improve energy efficiency [66] while promoting industrial structure optimization and technological innovation [67, 68], increasing the share of non-agricultural sectors, providing application scenarios for the digital economy, and accelerating the green transformation of traditional industries. Related policy support and regional coordination may further create favorable conditions for improving coordination between the two systems. These results agree with previous studies showing that new urbanization can improve environmental and energy efficiency through industrial restructuring and technological innovation [66–68]. This study complements previous provincial- and city-level research by showing that the relative contribution of urbanization to CCD varies across years.

4.4.3 Shapley Additive exPlanations dependency plot analysis

SHAP not only the analysis of global feature importance but also enables the examination of the dependence of SHAP values on individual feature values [51, 69]. As shown in Figure 12, the dependence plots illustrate the potentially nonlinear relationships between each selected factor and the predicted CCD and provide information on possible interactions between paired variables. The X-axis shows the feature values of the influencing factors, while the Y-axis shows the corresponding SHAP values. According to the analysis above, urbanization, government intervention, and scientific and technological innovation were relatively influential in most of the selected years. Therefore, SHAP dependence plots for these three features in 2011, 2015, 2019, and 2022 are presented.

It can be seen from Figure 12 that urbanization generally exhibits a positive but nonlinear association with the predicted CCD, particularly when its value exceeds a certain range in 2011, 2019, and 2022. By contrast, government intervention generally exhibits a negative association with the predicted CCD, although the relationship is not strictly monotonic in all years. Scientific and technological innovation also shows a nonlinear and time-varying association with the predicted CCD rather than a consistently increasing relationship. These patterns are broadly consistent with the SHAP summary results reported in Section 4.4.2, which showed that urbanization and scientific and technological innovation generally made positive contributions to the predicted CCD, whereas government intervention generally made a negative contribution.

The color of each point represents the value of another feature that may interact with the feature shown on the X-axis. For urbanization, government intervention is identified as the potential interaction feature in 2011, 2019, and 2022, whereas openness is identified in 2015. For government intervention, the potential interaction feature is industrial structure in 2011, urbanization in 2015, and openness in 2019 and 2022. Openness is identified as the potential interaction feature for scientific and technological innovation in all four years. These patterns suggest possible interactions among the selected features, but the dependence plots alone do not establish causal or synergistic effects. Therefore, further analysis based on SHAP interaction values or other interaction-effect methods is required to quantify the relationships between the selected feature pairs and their joint contributions to the predicted CCD. Such analysis could provide useful evidence for promoting the coordinated development of the digital economy and eco-efficiency.

5 Discussion

Using constructed indicator systems and analytical models, this study conducts an in-depth assessment of the CCD between the digital economy and eco-efficiency across 108 YREB cities and identifies regional development differences and the contributions of relevant influencing factors. The results show a general upward tendency in the CCD, but marked regional disparities remain. The downstream region has a higher CCD, possibly owing to its solid economic foundation and favorable policy support; the midstream region has made progress but at a relatively slower pace of development; and the upstream region has been upgrading more slowly, which may be related to its weak economic foundation and insufficient digital infrastructure. This spatial pattern points to regional differences in economic conditions, industrial composition, technological innovation capacity, and resource endowments.

From a temporal perspective, CCD increased relatively rapidly between 2011 and 2014, possibly reflecting the national emphasis on the YREB and related policy support. The growth rate slowed down from 2014 to 2017, with some cities showing slower progress in digital economy development and ecological governance. From 2017 onward, the CCD rapidly increased, coinciding with greater policy attention to ecological priorities and green development. However, after 2021, factors such as the complex global economic situation and recurring epidemics may have created difficulties for digital economic development and ecological governance in the YREB, leading to a slowdown in CCD growth.

The XGBoost–SHAP analysis reveals substantial differences in how the five explanatory variables are related to the model-predicted CCD. External openness is generally associated with higher predicted values, although the magnitude of this relationship changes over time. By contrast, greater government intervention tends to correspond

to lower CCD predictions, possibly because excessive administrative involvement may weaken regional innovation efficiency and firms' willingness to innovate. Industrial structure shows a comparatively small and predominantly negative effect, which may be attributable to the continued importance of energy-intensive industries and their constraints on the contribution of digitalization to eco-efficiency. Technological innovation and urbanization are generally linked to higher predicted CCD values, suggesting that both factors may facilitate greater synergy between digital economic development and ecological performance.

The observed inter-city variation in CCD may be attributable to disparities in urban governance capacity. Downstream megacities may have developed relatively mature digital urban governance systems: digital platforms are used to unify environmental monitoring, industrial supervision, and public service allocation, which may simultaneously advance digital economic development and improve eco-efficiency. Midstream provincial capitals have partially built digital governance platforms, but cross-department data-sharing barriers may restrict the coordinated governance of digital and ecological systems. Upstream small and medium-sized cities may lack unified urban digital operation platforms, while fragmented ecological governance and underdeveloped digital infrastructure may jointly restrict the coordination of the two systems. Therefore, narrowing CCD gaps essentially requires optimizing differentiated urban digital and ecological governance schemes.

In summary, despite the upward trend in the CCD between the digital economy and eco-efficiency across the YREB, substantial regional disparities persist. Future efforts should prioritize stronger interregional coordination, more effective policy support, enhanced regional innovation capacity, and the upgrading of industrial structures to reconcile high-quality economic development with ecological conservation.

6 Conclusions and Recommendations

6.1 Conclusions

Using panel data for 108 YREB cities from 2011 to 2022, this study constructs the digital economy index with the EWM and estimates eco-efficiency using the Super-EBM model. The CCD framework is then combined with XGBoost to examine the spatio-temporal evolution of coordination between the two systems and identify its principal associated factors. The key findings are presented below:

(1) The digital economy followed an overall upward trajectory, although its growth rate varied over time. It increased steadily between 2011 and 2017, accelerated markedly from 2017 to 2021, and then recorded a slight decline in 2022. In terms of spatial distribution, digital economy development across the YREB follows a clear “downstream > midstream > upstream” pattern. The eco-efficiency demonstrates a gentle fluctuation tendency of “rising–falling–rising again–falling again–rising again,” with a W-shaped trend from 2016 to 2022. Regarding spatial characteristics, there are regional variations in the YREB, generally showing the pattern of “upstream > midstream > downstream.”

(2) CCD shows a generally stable upward trend. Temporally, from 2011 to 2014, the CCD increased rapidly, possibly reflecting the national emphasis on the digital economy and ecological protection; from 2014 to 2017, the growth rate slowed down; after 2017, following the strengthened emphasis on ecological priority and green development, the CCD increased rapidly. Spatially, CCD shows marked spatial heterogeneity of “downstream > midstream > upstream,” with the downstream area having a higher degree of coupling coordination, possibly owing to its superior economic foundation and earlier development advantages, followed by the midstream, while the upstream area has improved more slowly, possibly because of its weak economic foundation and insufficient digital infrastructure.

(3) The XGBoost–SHAP results show that external openness and technological innovation are generally associated with higher predicted CCD values, while government intervention is associated with lower values. Industrial structure exhibits a relatively limited and predominantly negative contribution. Urbanization was the most influential factor in 2011, 2019, and 2022, while openness ranked first in 2015. The SHAP dependence plots further indicate nonlinear and time-varying relationships between the selected influencing factors and the predicted CCD.

6.2 Recommendations

To support more balanced progress in digital economic development and eco-efficiency across the YREB, this study proposes the following general and region-specific policy measures.

(i) Strengthen interregional cooperation and narrow the digital divide. In the downstream regions, for instance, in the Yangtze River Delta region, the digital economy develops more swiftly, while the midstream and upstream regions are lagging behind. It is recommended to strengthen cooperation among the upstream, midstream, and downstream regions and to aid the central and western regions in enhancing their digital infrastructure via technology sharing and the transfer of industries.

(ii) Reinforce policy support for regional innovation. Governments should adopt targeted incentives, including tax relief, research funding, and talent-support programs, to attract skilled professionals and innovative enterprises to the central and western regions. Public investment in digital infrastructure should also be increased. Meanwhile,

greater support for green technological innovation could accelerate the intelligent and low-carbon upgrading of traditional industries.

(iii) Integrate green objectives into digital development strategies. Local governments should coordinate digital economy policies with ecological protection and emission-reduction targets. For example, big-data platforms could be used to monitor corporate carbon emissions in real time, strengthen environmental supervision, and facilitate the low-carbon transformation of the real economy. Authorities should also promote cleaner production, reward enterprises with strong environmental performance, and encourage residents to adopt low-carbon consumption patterns.

(iv) Formulate development strategies suited to regional conditions. Each subregion should design policies according to its location, resource endowments, and industrial foundation. Downstream areas could prioritize high-technology industries, low-carbon hydrogen and photovoltaic technologies, and ecological demonstration zones. Midstream and upstream provinces, including Sichuan and Yunnan, could make greater use of their hydropower and solar resources to establish clean-energy bases, reduce dependence on fossil-fuel extraction, and better balance economic growth with ecological conservation.

(v) Expand energy-efficient digital infrastructure. Investment in digital infrastructure should be accelerated in key growth areas, including the Yangtze River Delta and the Chengdu–Chongqing Economic Circle. At the same time, renewable energy and energy-efficient technologies should be incorporated into the construction and operation of data centers, communication networks, and other digital facilities to limit the additional energy demand generated by digital economic expansion.

In addition to these general measures, differentiated actions are required for the downstream, midstream, and upstream regions.

(vi) Downstream regions: advance high-end digital industries and low-carbon demonstration projects. Downstream cities should accelerate the research, application, and commercialization of low-carbon hydrogen and photovoltaic technologies. Integrated digital platforms for cross-city ecological governance should also be established. Further expansion of energy-intensive industries should be restricted, while digital technologies and green management experience should be transferred to the middle and upper reaches through industrial cooperation and technical assistance.

(vii) Midstream regions: promote the green and digital upgrading of traditional industries. Midstream areas should accelerate the digital transformation of manufacturing while enforcing stricter industrial emission standards. Provincial innovation-sharing platforms and cross-boundary ecological governance mechanisms should be developed. These regions may also receive low-carbon digital industries transferred from downstream areas, but should avoid accepting highly polluting or technologically outdated production capacity.

(viii) Upstream regions: improve digital connectivity and develop clean-energy and ecological industries. Greater fiscal support should be directed toward basic digital networks in mountainous and small- and medium-sized cities. Upstream areas should capitalize on hydropower and photovoltaic resources to develop clean-energy industries, while prioritizing ecological conservation and supporting low-pollution sectors such as ecotourism and digital agriculture. Talent-attraction policies should be strengthened to reduce the loss of skilled workers, and the entry of carbon-intensive heavy industries should be strictly controlled.

6.3 Limitations

Despite offering new insights into the spatial and temporal evolution of the CCD between the digital economy and eco-efficiency in the YREB and the factors associated with it, the study has several limitations. First, data are missing for some cities within the YREB, which, to some extent, undermines the comprehensiveness and accuracy of the study. Second, the Super-EBM and XGBoost models utilized in the study can quantify eco-efficiency and assess the factors influencing CCD, but they still have some limitations in explaining specific causal relationships, and the effects of some factors may require more detailed empirical tests. Third, the time span of the study is 2011–2022, which fails to cover earlier or later periods and thus may not be able to present a complete picture of the development trend of the YREB over a longer historical period.

Author Contributions

Conceptualization, J.W.T.; methodology, J.S.; software, H.L.Y.; formal analysis, K.H.L.; investigation, Y.Y.; resources, J.W.T.; data curation, P.L.D.; writing—original draft preparation, D.L.M.; writing—review and editing, D.L.M.; visualization, K.H.L.; supervision, J.S.; project administration, R.N.C.; funding acquisition, H.L.Y.; validation, C.H. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data supporting the findings of this study are subject to restrictions and cannot be shared publicly. Requests for access may be directed to the corresponding author, subject to permission from the data provider.

Conflicts of Interest

The authors declare no conflicts of interest.

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